



<http://www.eirene.de>

A quick and easy way to learn
EIRENE



<http://www.eirene.de>

There is no!

Why is EIRENE so complicated??

Because the problem it solves is so complicated!!



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General Strategy:

Model any particular physics effect
as simple as possible.

But not simpler.

(A.Einstein)



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Before we start

EIRENE is a Monte Carlo Code developed and occasionally extended for particular physical topics that interested us and others in the past 20 years.

It **is not a commercial tool**, it must not be confused with such.

It is part of our ongoing research work on atomic processes in fusion plasmas, it therefore **is a research project**, and a moving target.



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Documentation

Our laboratory notes about the EIRENE code have been edited and made available in form of reports of FZ- Jülich **JUEL 1974 (Aug 1984)** and **JUEL 2599 (March 1992)** (both sold out).

EIRENE has been linked to a number of external codes (SOLXY, TRANSP, DIVIMP, B2, EMC3, OSM, currently: B2.5, FIDAP, E3D).

Our notes on interfacing B2 and EIRENE have been turned into reports on the various KFA-NET contracts 1986 --1990, and, in edited form, again been published as FZ- Jülich report **JUEL-2872 (March 1994)**.

The material kept rapidly changing. In order to keep the documentation up to date and for assisting communication between people working on/with EIRENE or B2-EIRENE the material was uploaded onto a web-domain: [www:eirene.de](http://www.eirene.de).

It was password protected because, as with the code itself, it needed to be verified first that any particular user of EIRENE does

not confuse these notes with a „professional code documentation“.

In May 2001 the password protection was removed, it is now generally accessible.



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EIRENE HISTORY

EIRENE and **DEGAS** have been developed, independently, from the **AURORA** code (Hughes, Post, 1979), in the early eighties. They are brother and sister.

NIMBUS was derived from a neutronics code.

Synchronisation between **NIMBUS** and **EIRENE** was achieved during heavy drinking sessions in the Ponchbowl pub, Abingdon between A. D'Mattheis and D.Reiter (1985-1986)



- Domain discretisation by surfaces, not cells
 - very easy to construct complicated geometries
 - very difficult to know cell volumes (see NLERG-option)
- 6 different geometry levels, from 1D to 3D
- All “events” (collision processes) externally prescribed
 - Extremely flexible to implement new processes
 - The user must, actually, know a lot about atomic processes
- 3 estimators: collision counting (DEGAS)
pathlength (EIRENE-default)
exponential transform (NIMBUS)
- Statistical error bars for all tallies
- User defined tallies (weighting of distribution)
see section 3.2 of manual

LEV GEO=1: Cartesian, 1D, 2D, 3D
basically for testing, and 1-speed transport theory

* SLAB MODEL FOR TEST OF DIFFUSION APPROXIMATION

SCALING FACTORS

FACT-X= 7.500E+01

FACT-Y= 7.500E+01

ORIGIN

CHXO= 5.000E+01

CHYO= 5.000E+01

PLOTTED AT

Z = 0.000E+00

* LOCATE(1)

* ELECTR. IMPACT(2)

○ ELASTIC COLL.(3)

x CHARGE EXCHANGE(4)

□ FOKKER-PLANCK(5)

▲ SURFACE(6)

◆ SPLITTING(7)

▼ RUSSIAN ROULETTE(8)

► PERIODICITY(9)

► RESTART:A. SPLIT(10)

► SAVE.COND. EXP.(11)

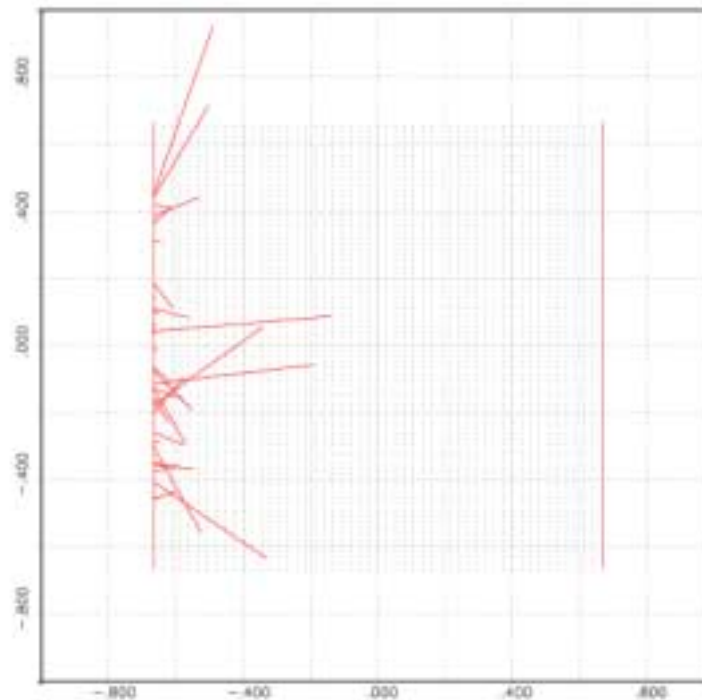
► RESTART.COND EXP(12)

△ TIME LIMIT(13)

○ GENERATION LIMIT(14)

● FLUID LIMIT(15)

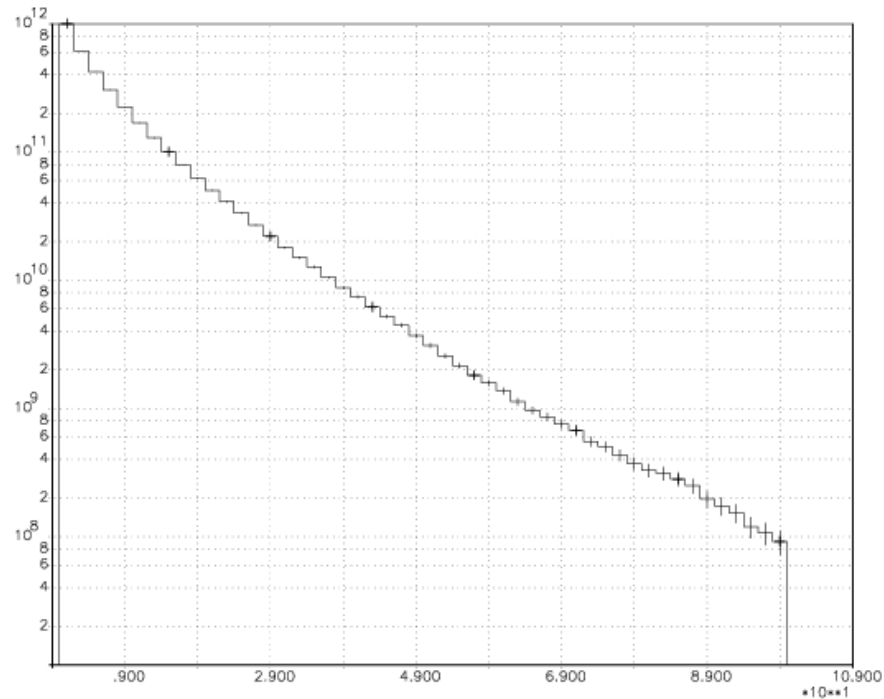
□ ERROR DETECTED



Example: Test of validity of diffusion models

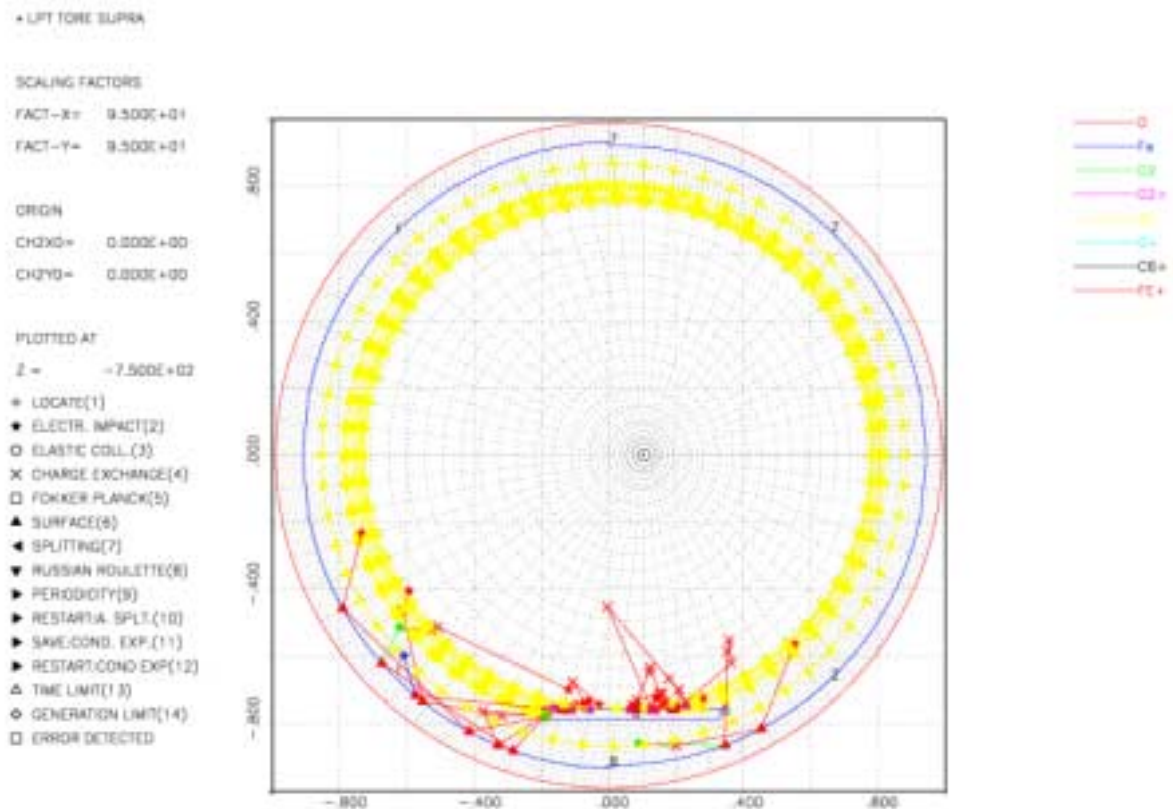
here: 1D density profile (incl. stat. error bars)

* SLAB MODEL FOR TEST OF DIFFUSION APPROXIMATION
DEFAULT VOLUME AVERAGED TALLY, TRACKLENGTH ESTIMATED
MIN. ABSZ/ISSA = 0.000E+00
MAX. ABSZ/ISSA = 1.000E+02
+ PARTICLE DENSITY (ATOMS)
H
CM**--3
MAX. VALUE 1.000E+12
MIN. VALUE 9.152E+07



LEV GEO=2: 1D, 2D polar coordinates

here CIEL pumplimiter included, collisionless ions provide spatial distribution of recycling source



LEV GEO=3: structured grid for CFD codes (e.g. B2)

cylindrical or toroidal symmetry available

• COUPLING DIRECT TO BRAHMS CODE: ASSESS UPGRADE SINGLE-NULL

SCALING FACTORS

FACT-X= 1.500E+02

FACT-Y= 1.500E+02

ORIGIN

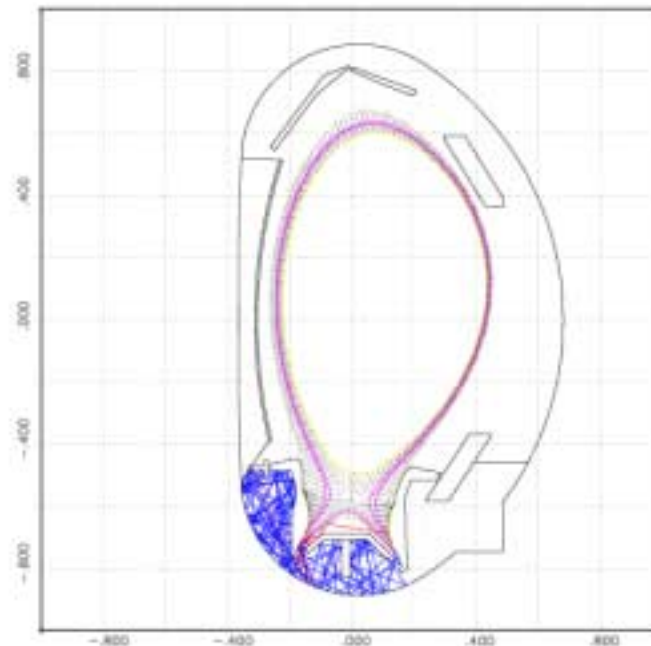
CH2X= 1.500E+02

CH2Y= 0.000E+00

PLOTTED AT

Z = 0.000E+00

- + LOCATE(1)
- ★ ELECT. IMPACT(2)
- ELASTIC COLL.(3)
- × CHARGE EXCHANGE(4)
- FOKKER-PLANCK(5)
- ▲ SURFACE(6)
- ◆ SPLITTING(7)
- ▼ RUSSIAN ROULETTE(8)
- PERIODICITY(9)
- RESTART-A: SPLIT(10)
- SAVE COND. EXP.(11)
- RESTART-COND EXP(12)
- △ TIME LIMIT(13)
- ◇ GENERATION LIMIT(14)
- ★ FLUID LIMIT(15)
- ERROR DETECTED



LEV GEO=4 Finite element discretisation, 2D unstructured grids.

used for “vacuum region” outside B2 grid in non-linear MC applications and, in the past, for coupling to FEM edge codes

* EIRENE input file generated with uinp v. 2.0

SCALING FACTORS

FACT-X= 6.000E+01

FACT-Y= 6.000E+01

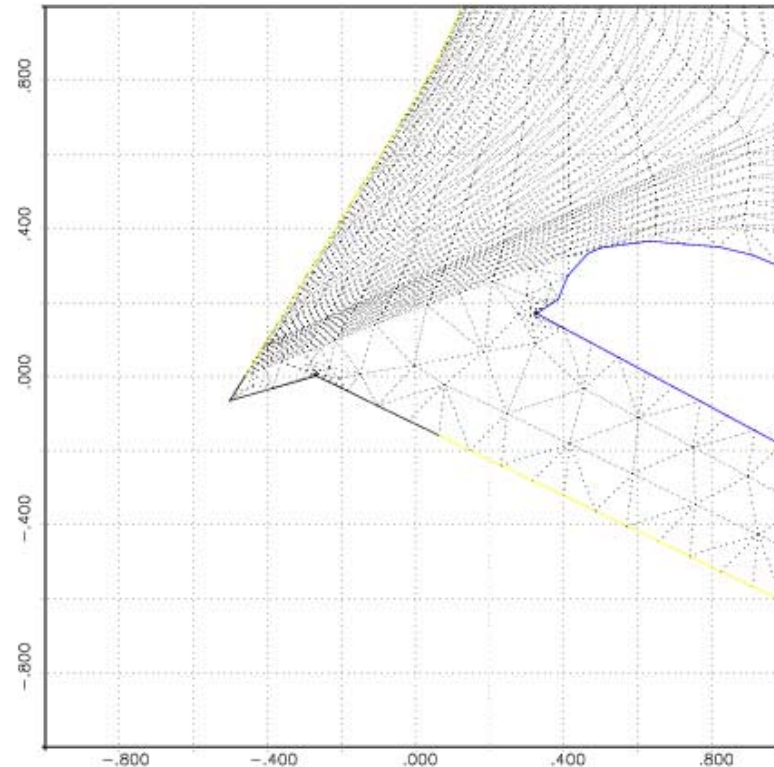
ORIGIN

CH2X0= 4.372E+02

CH2Y0= -3.809E+02

PLOTTED AT

Z = 0.000E+00



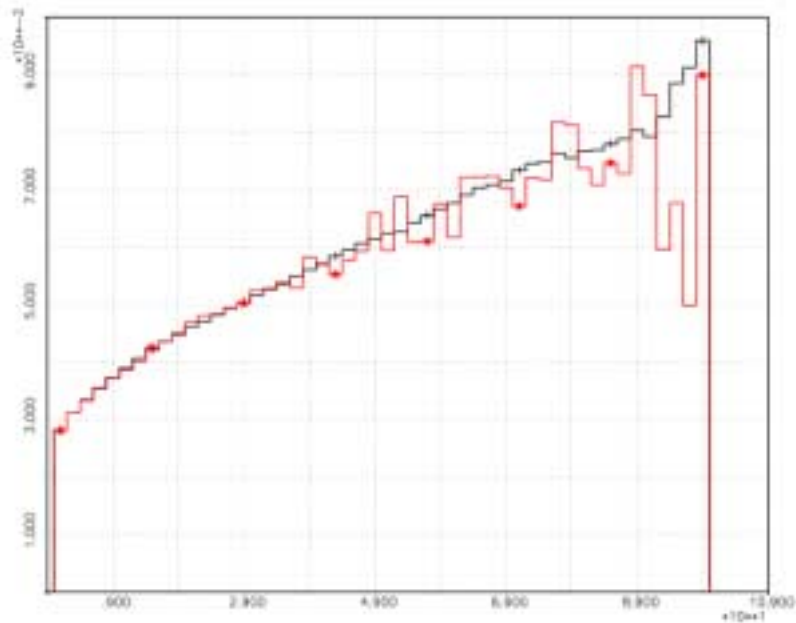
Comparison: tracklength vs. collision estimator

here: atom temperature profile, 1D LEVGEO=1

+ SLAB WOOD, FOR TEST OF DIFFUSION APPROXIMATION
ALGORITHM: FUNCTION OF VOLUME AVERAGED VALUES
MIN_ABSZ_GSA = 0.000E+00
MAX_ABSZ_GSA = 1.000E+02

+ kinetic temperature, tracklet
in
eV
MAX VALUE 3.588E-02
MIN VALUE 2.811E-02

• kinetic temperature, celllet
in
eV
MAX VALUE 3.143E-02
MIN VALUE 2.814E-02





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There are two kinds of codes

- categorical: **easy to use**, very detailed, specialised to one **specific** case
- general: **hard to use** and document, many applications, must contain as little as possible hard wired physics in order to **retain flexibility**

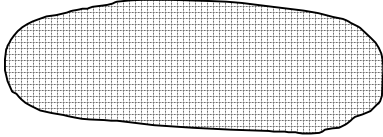
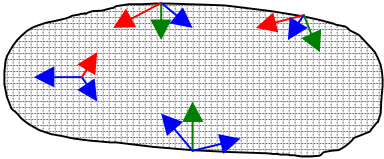
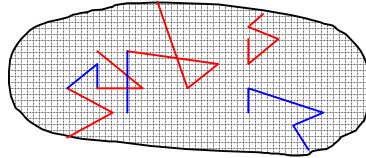
Many people have suffered a lot because of the complexity of the EIRENE input.

But the retained flexibility has enabled others to study physics problems not anticipated when EIRENE was originally written.

Next: example photon transport, S.Wiesen, 6 month

EIRENE

A general purpose linear transport
(Boltzmann equation) solver

DOMAIN	(discretised)	
SOURCE	of Objects	
OBJECTS	move on straight lines between Events	
EVENTS	Change direction and / or species at these events or their history ends at these events ("SINK for Objects")	

By a stochastic simulation of these random walks
a solution to the Boltzmann equation is found

	Neutral gas transport	Radiation transfer
DOMAIN:	Fusion edge plasma region (outer zone of the plasma)	Stellar atmospheres, high pressure gas discharges,.....
OBJECTS:	neutral atoms H (isotopes) neutral molecules H ₂ (isotopomeres) short living molec.Ions H ₂ ⁺ (isotopomeres)	Photons
SOURCE:	gas puff recycling at surfaces volume recombination	Spontaneous line emission, induced line emission
SINK:	ionisation pumping (at surfaces)	Photon absorption, Escape from domain
EVENTS:	neutral-plasma collisions (↑ A & M data) reflection at surfaces (↑ surface data)	Photon scattering Reflection at surfaces (mirror)

Kinetic (transport) equation

- for particles travelling in straight lines between collisions
- with no forces acting on them between collisions

$$f(\vec{r}, \vec{v}, t)$$

$$f(\vec{r}, E, \vec{\Omega}, t)$$

$$\begin{aligned} \frac{\partial f(E, \vec{\Omega})}{\partial t} + v \vec{\Omega} \cdot \nabla f(E, \vec{\Omega}) &= S(E, \vec{\Omega}) - v \sigma_a(E) f(E, \vec{\Omega}) \\ &+ \int_0^\infty dE' \int_{4\pi} d\vec{\Omega}' \left[v' \sigma_s(E' \rightarrow E, \vec{\Omega}' \cdot \vec{\Omega}) f(E', \vec{\Omega}') - v \sigma_s(E \rightarrow E', \vec{\Omega} \cdot \vec{\Omega}') f(E, \vec{\Omega}) \right] \end{aligned}$$

This equation is algebraically very complex,
but it has a very simple physical content
(conservation in phase space)

There are numerous applications:

- neutron migration in nuclear reactors
- radiative transfer
- neutrino flow in astrophysics
- neutral particle transport in plasmas
- Knudsen flow
- gamma-ray transport in shielding studies
-

Use scalar transport flux as dependent variable
(rather than distribution function $\mathbf{v} \cdot \mathbf{f} = \psi$)

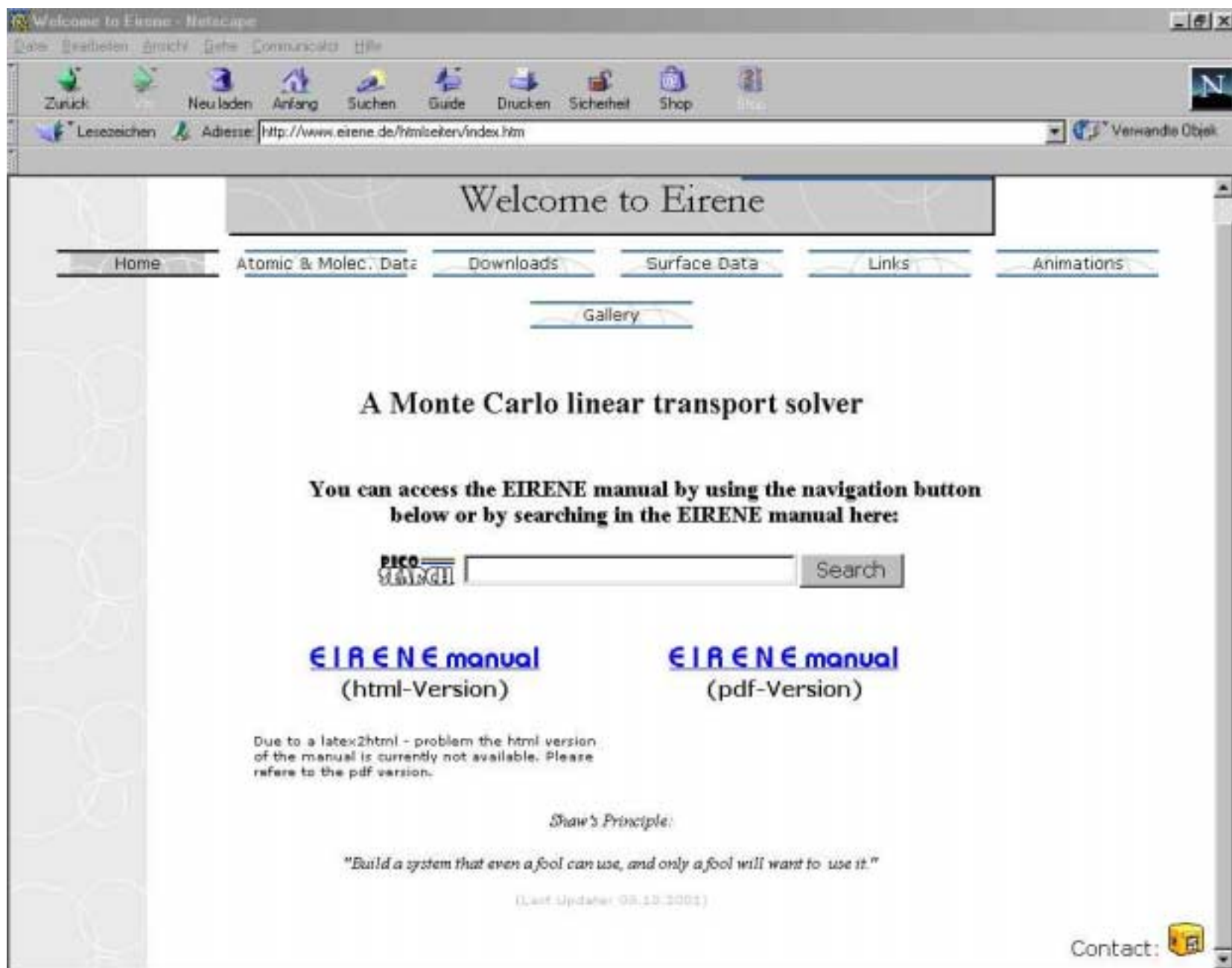
$$\frac{1}{v} \frac{\partial \psi(E, \vec{U})}{\partial t} + \vec{U} \cdot \nabla \psi(E, \vec{U}) = S(E, \vec{U}) - \sigma_a(E) \psi(E, \vec{U}) \\ + \int_0^\infty dE' \int_{4\pi} d\vec{U}' [\sigma_s(E' \rightarrow E, \vec{U}' \cdot \vec{U}) \psi(E', \vec{U}') - \sigma_s(E \rightarrow E', \vec{U} \cdot \vec{U}') \psi(E, \vec{U})]$$

Boundary condition prescribe: ingoing flux

$$\psi(\vec{r}_s, E, \vec{\Omega}, t) = \Gamma(\vec{r}_s, E, \vec{\Omega}, t) \quad \vec{n} \cdot \vec{\Omega} < 0$$

$\vec{n}$: outer surface normal

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Simple transformations of variables:

$$\nu \Rightarrow c, E \Rightarrow \nu, \psi \Rightarrow I = h \nu c f$$

Equation of radiation transfer

$$\frac{1}{c} \frac{\partial I(\nu, \vec{\Omega})}{\partial t} + \vec{\Omega} \cdot \nabla I(\nu, \vec{\Omega}) + \sigma_a(\nu) I(\nu, \vec{\Omega}) = q(\nu) + \int_0^\infty d\nu' \int_{4\pi} d\vec{\Omega}' \frac{\nu}{\nu'} \sigma_s(\nu' \rightarrow \nu, \vec{\Omega}' \cdot \vec{\Omega}) I(\nu', \vec{\Omega}') - \int_0^\infty d\nu' \int_{4\pi} d\vec{\Omega}' \sigma_s(\nu \rightarrow \nu', \vec{\Omega} \cdot \vec{\Omega}') I(\nu, \vec{\Omega})$$

Complication: Quantum statistics : Photons are Bosons

hence: emission and scattering are enhanced by the number of photons already in the final state.

(“induced processes”, or “stimulated processes”)

$$\begin{aligned} \frac{1}{c} \frac{\partial I(\nu, \vec{\Omega})}{\partial t} + \vec{\Omega} \cdot \nabla I(\nu, \vec{\Omega}) &= q(\nu) \left[1 + \frac{c^2 I(\nu, \vec{\Omega})}{2h\nu^3} \right] - \sigma_a(\nu) I(\nu, \vec{\Omega}) \\ &+ \int_0^\infty d\nu' \int_{4\pi} d\vec{\Omega}' \frac{\nu}{\nu'} \sigma_s(\nu' \rightarrow \nu, \vec{\Omega}' \cdot \vec{\Omega}) I(\nu', \vec{\Omega}') \left[1 + \frac{c^2 I(\nu, \vec{\Omega})}{2h\nu^3} \right] - \\ &- \int_0^\infty d\nu' \int_{4\pi} d\vec{\Omega}' \sigma_s(\nu \rightarrow \nu', \vec{\Omega} \cdot \vec{\Omega}') I(\nu, \vec{\Omega}) \left[1 + \frac{c^2 I(\nu', \vec{\Omega}')}{2h\nu'^3} \right] \end{aligned}$$

In our current applications (excluding astrophysical applications for now):

σ_s very small

hence: stimulated scattering very small

hence:

despite stimulated emission:

radiation transfer equation remains linear,

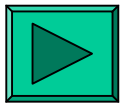
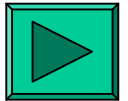
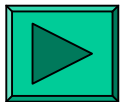
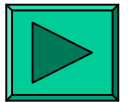
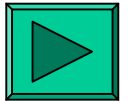
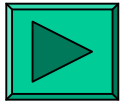
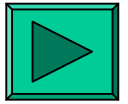
if upper and lower state populations are given (this latter fact requires iteration)

We must solve a three state nonlinear problem:

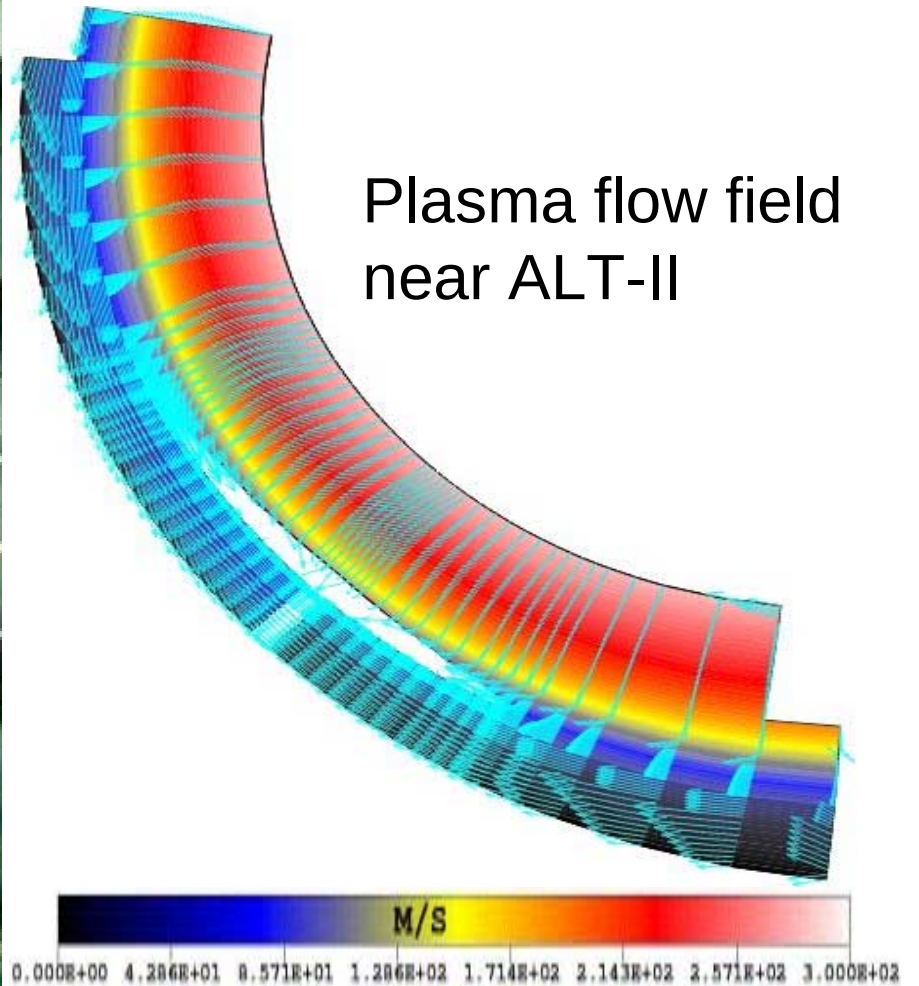
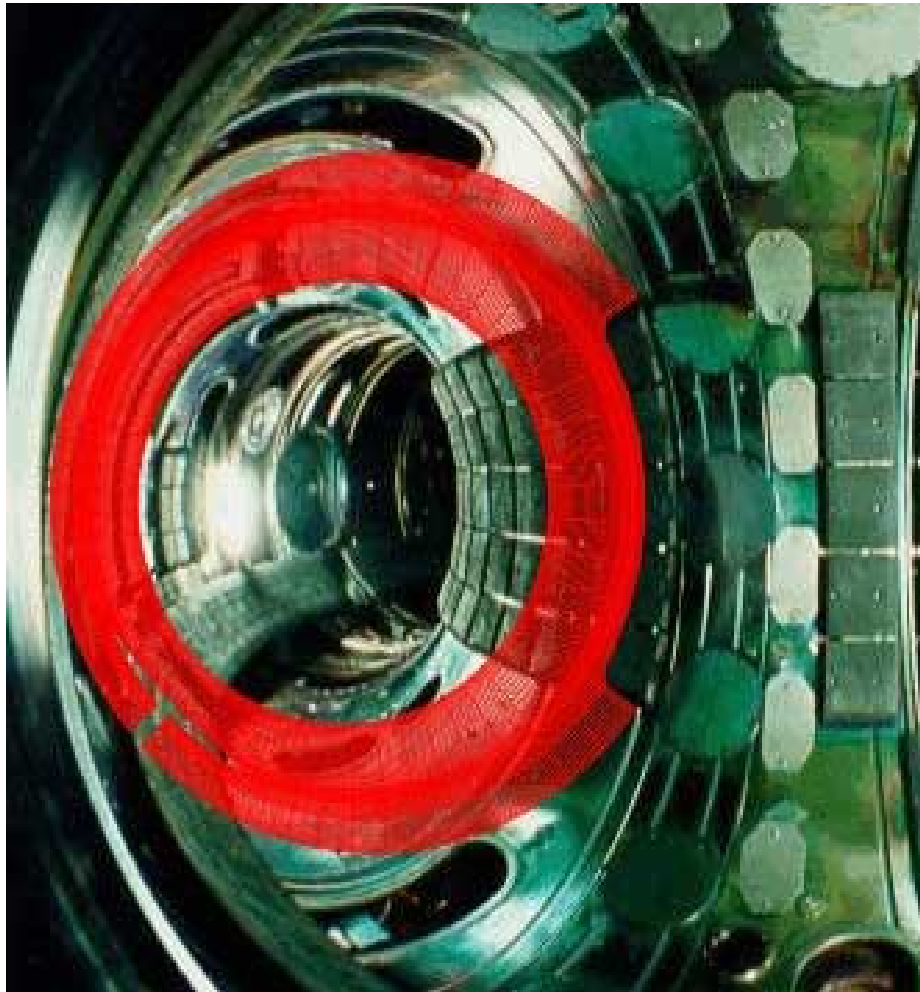
n_1, n_2, I

Required extensions of Eirene

- Phase space transformation:
- Line shape profiles (broadening)
- Test simple cases which allow numerical solution
- Planck distribution as limiting case
- General 3 dimensional configurations
- Nonlinear iterations for non-LTE conditions
- Deal with stiffness



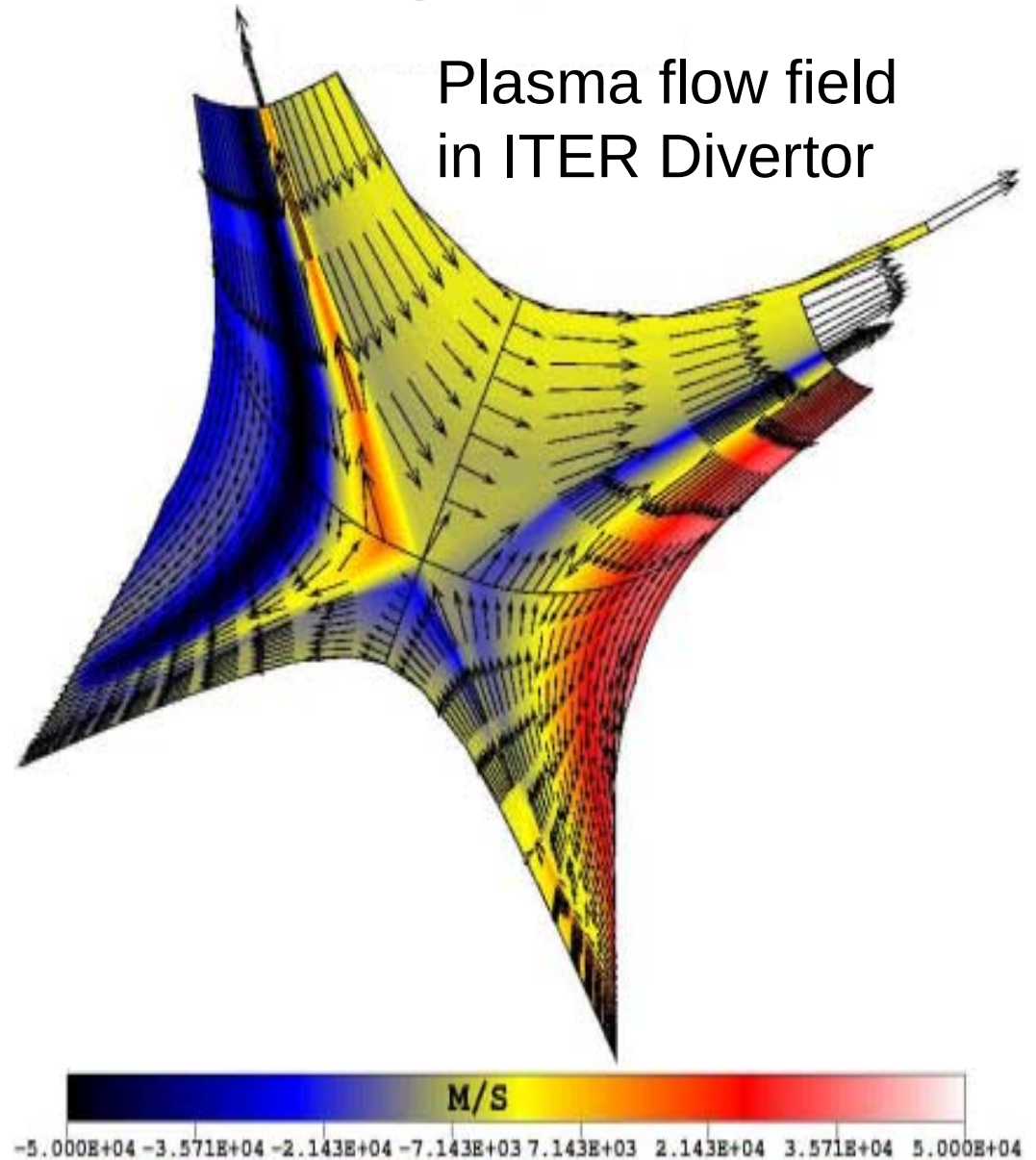
B2-EIRENE-Simulationsrechnung für TEXTOR



B2-EIRENE-Simulationsrechnung für ITER FEAT



Plasma flow field
in ITER Divertor



ITER-FEAT (B2-EIRENE simulation)



X MIN = 4.07E+02
MAX = 6.52E+02
Y MIN = -4.59E+02
MAX = -2.61E+02
Z MIN = 0.00E+00
MAX = 0.00E+00

FZ-Juelich

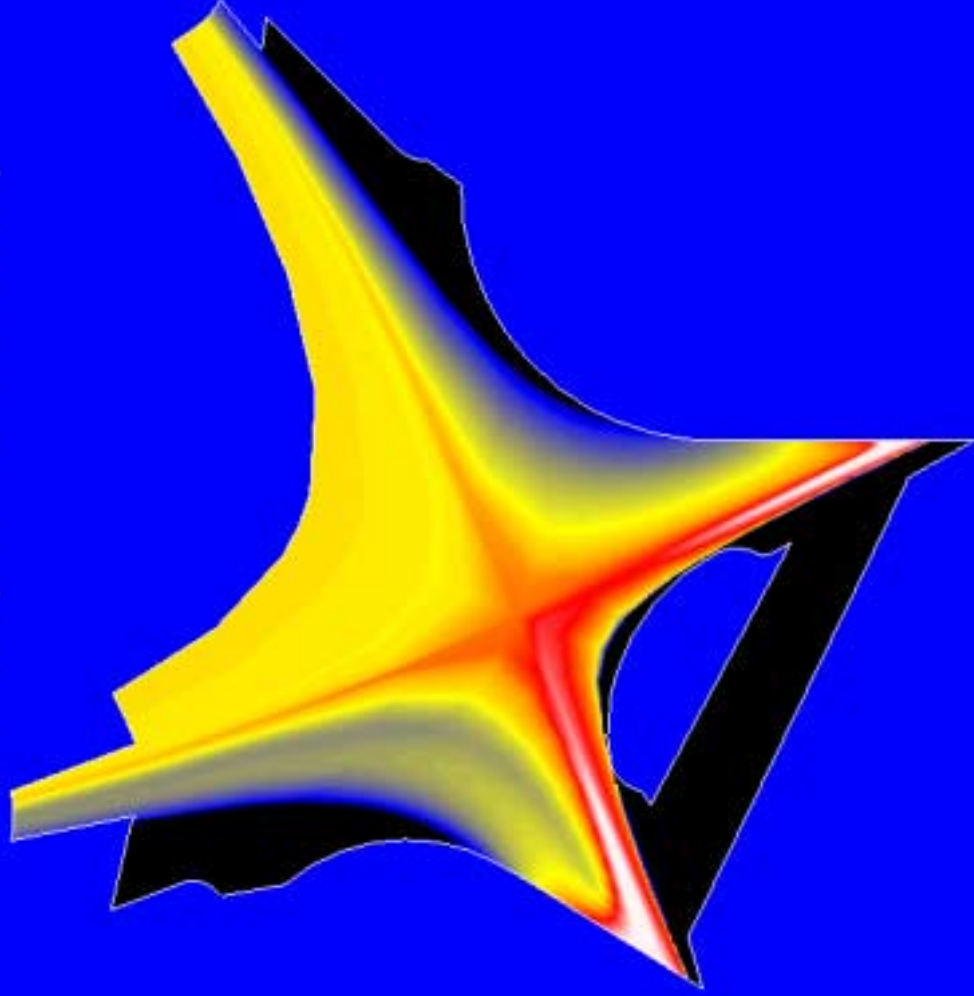
Electron Temperature

R A P S

eV

0.000E+00 1.429E+01 2.857E+01 4.286E+01 5.714E+01 7.143E+01 8.571E+01 1.000E+02

ITER-FEAT (B2-EIRENE simulation)



X MIN = 4.07E+02 Y MIN = -4.59E+02 Z MIN = 0.00E+00
 MAX = 6.74E+02 MAX = -1.97E+02 MAX = 0.00E+00

FZ-Juelich

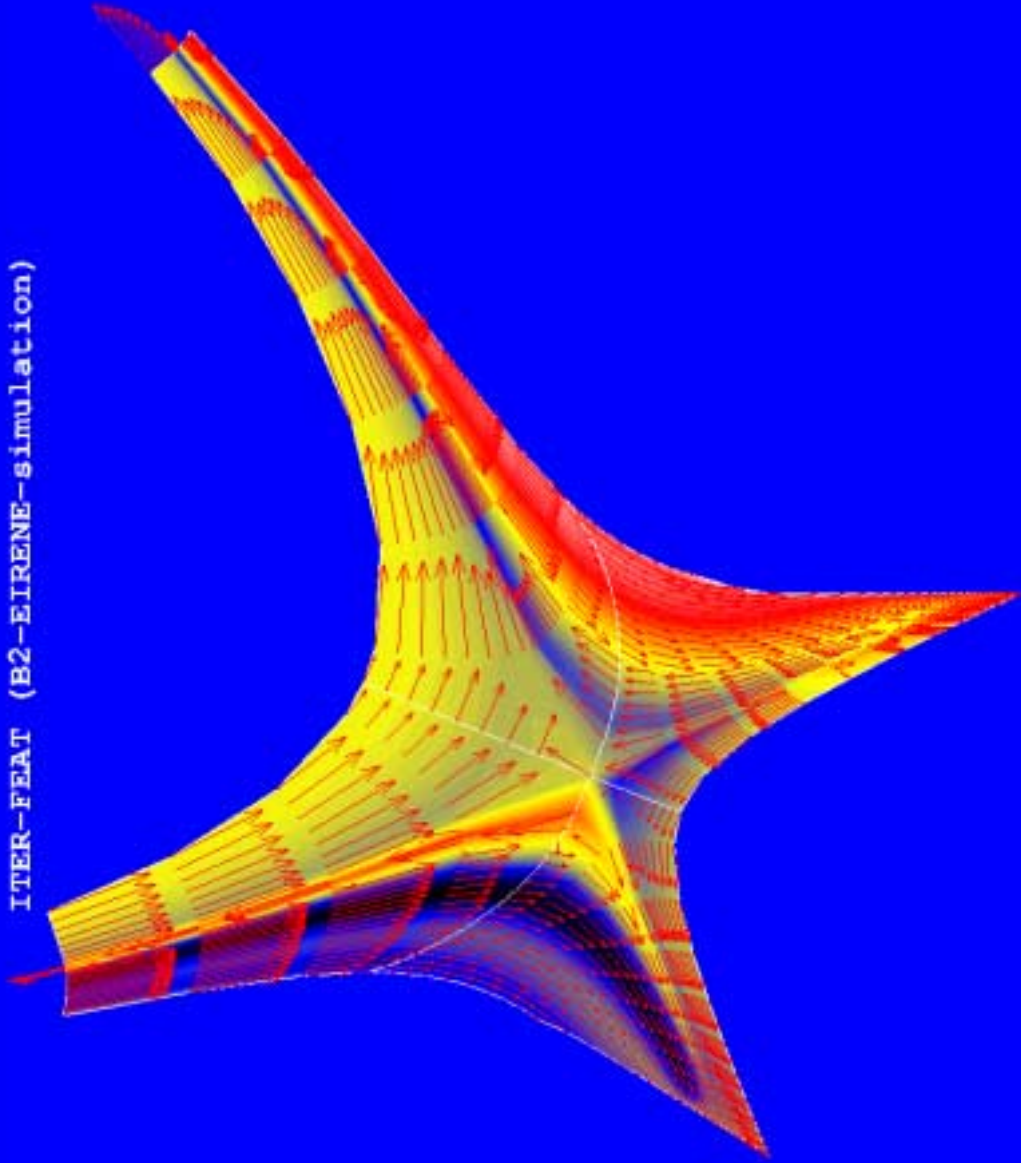
Electron Density (log scale)

R A P S

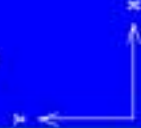
CM**--3

1.200E+01 1.243E+01 1.286E+01 1.329E+01 1.371E+01 1.414E+01 1.457E+01 1.500E+01

ITER-FEAT (B2-EIRENE-simulation)



VERZ. FAKT. : 6.10E-02



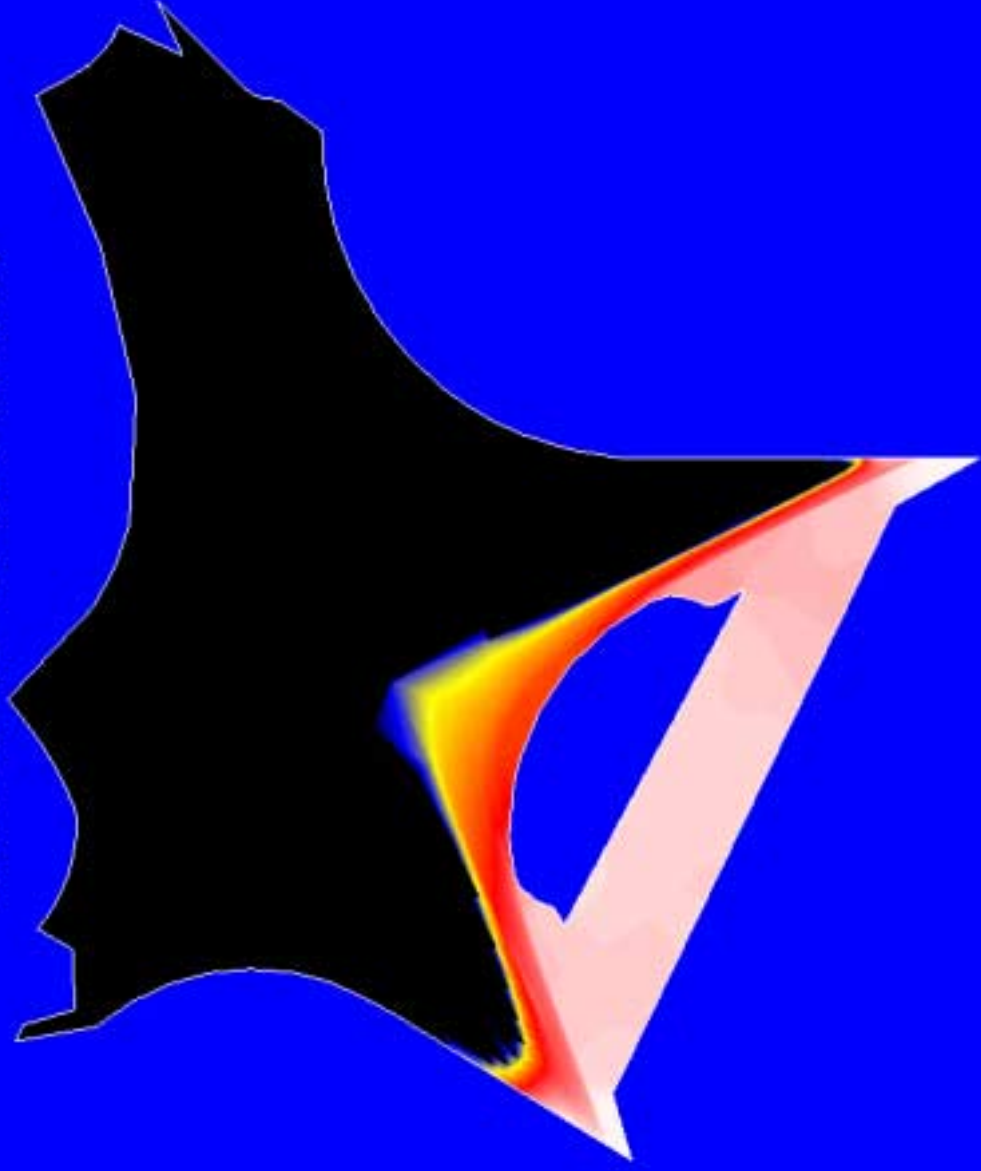
x MIN = 0.00E+00
MAX = 0.00E+00

y MIN = -4.46E+02
MAX = -1.93E+02

z MIN = 4.10E+02
MAX = 7.02E+02

FZ-Juelich		Plasma flow field		R A P S			
		cm/s					
-5.040E+04	-3.620E+04	-2.201E+04	-7.815E+03	6.379E+03	2.057E+04	3.477E+04	4.896E+04

ITER-FEAT (B2-EIRENE simulation)



Y
X
-90.0
0.0
90.0

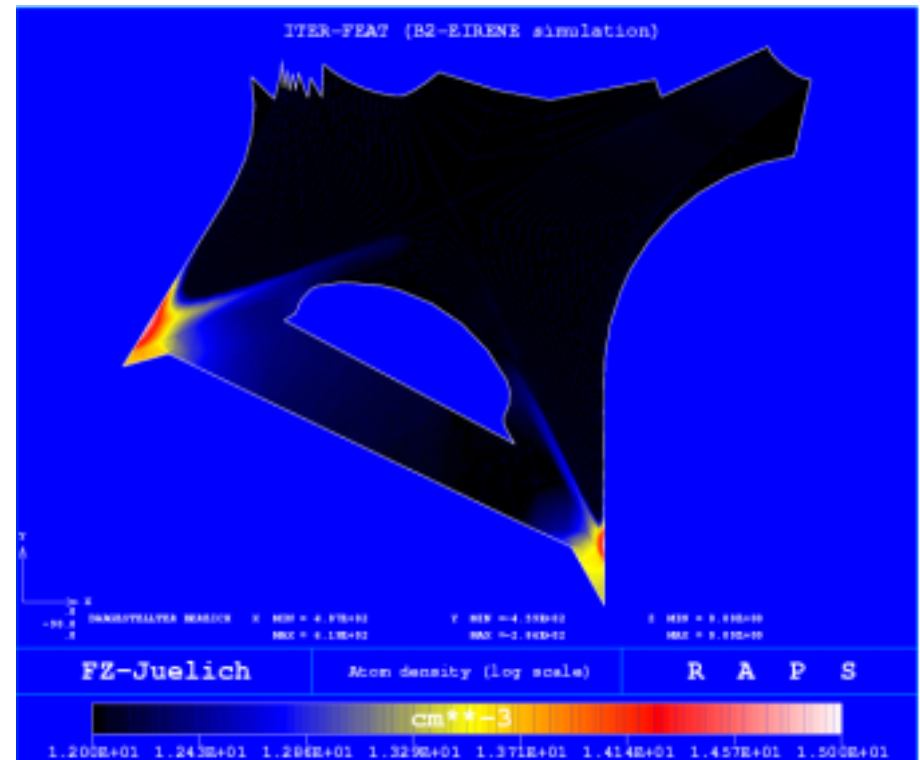
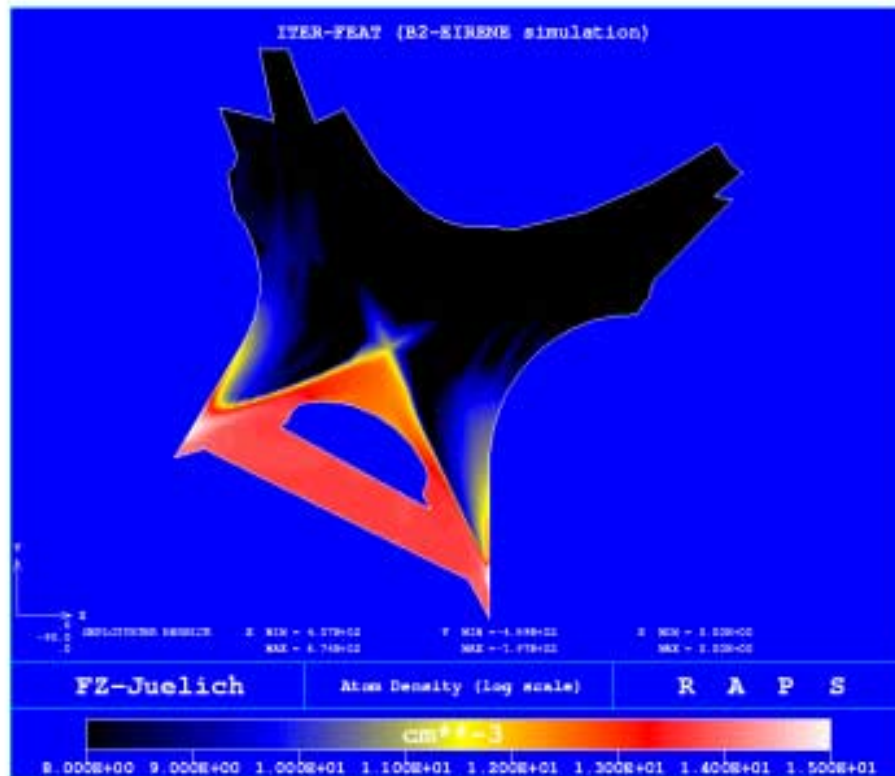
Z MIN = 0.00E+00
MAX = 0.00E+00

Y MIN = -4.59E+02
MAX = -2.87E+02

X MIN = 4.07E+02
MAX = 6.82E+02

FZ-Juelich		Molecule Density (log scale)		R A P S	
8.000E+00	9.000E+00	1.000E+01	1.100E+01	1.200E+01	1.300E+01
1.400E+01	1.500E+01				

CM**-3



Effects of radiation transfer:

- drastic reduction in atom density
- plasma shifts from recombining towards ionising state

Plasma response to new divertor state not yet studied

Current plasma edge simulations are either

too optimistic (optically thin case)

or

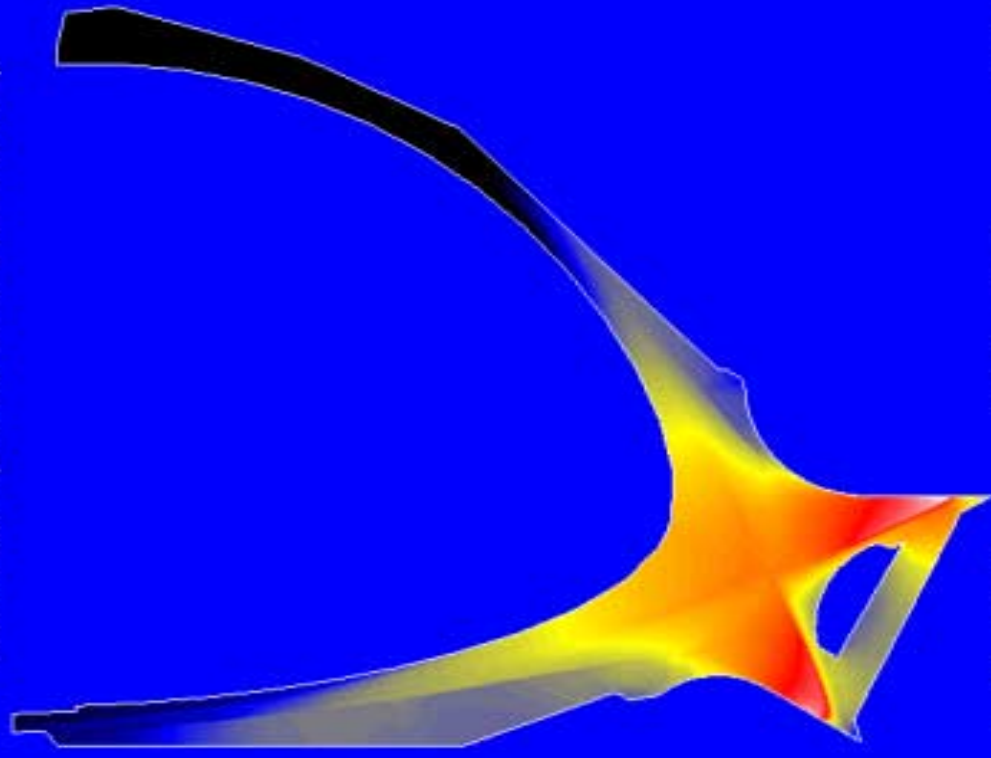
too pessimistic (optically thick case)

or

too crude (escape factors based upon simple estimates (ADAS))

now: full radiation transfer calculation

ITER-FEAT (B2-EIRENE simulation)



Z MIN = 0.00E+00
MAX = 0.00E+00

Y MIN = -4.53E+02
MAX = 3.02E+01

X MIN = 4.05E+02
MAX = 8.50E+02

FZ-Juelich

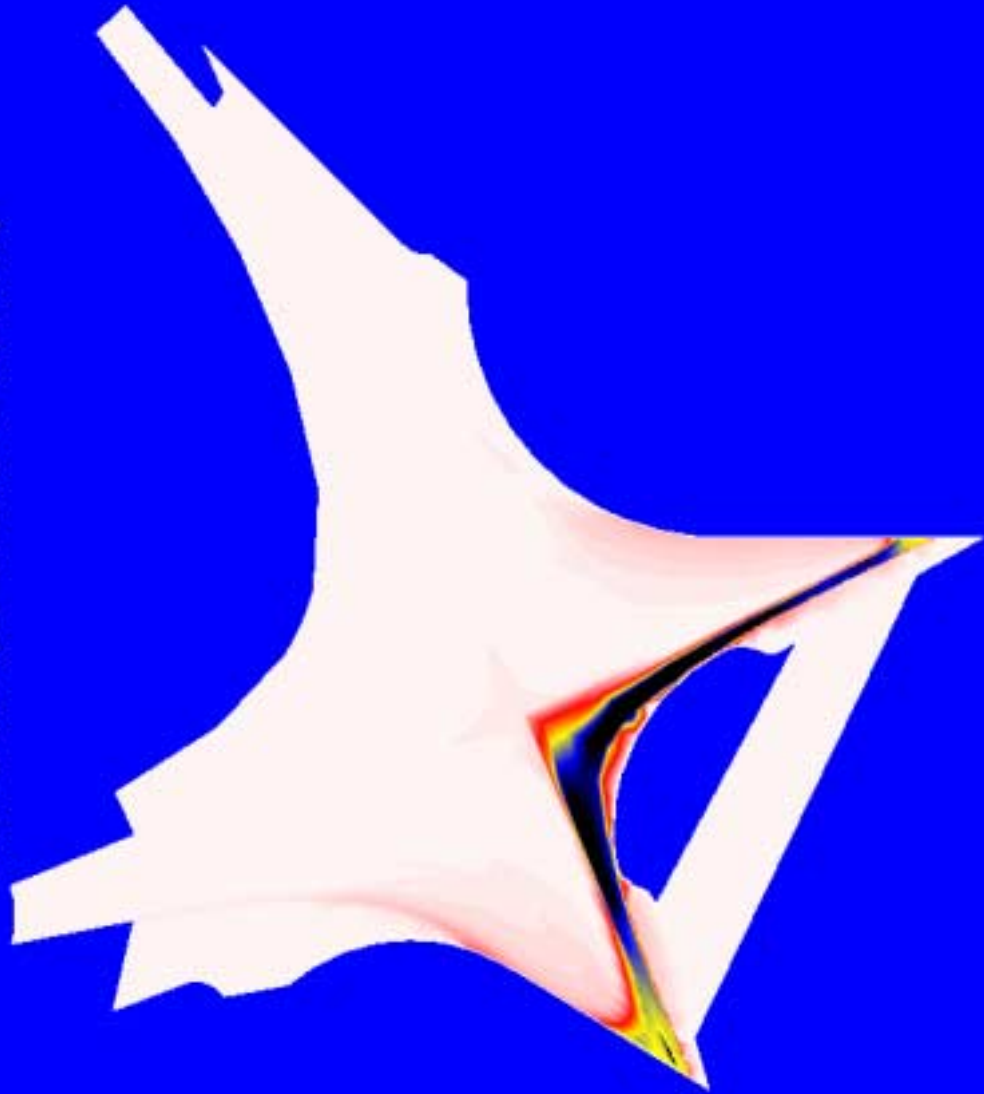
Lyman alpha photon density

R A P S

$\text{cm}^{-3} \times 10^{-3}$

6.000E+00 6.429E+00 6.857E+00 7.286E+00 7.714E+00 8.143E+00 8.571E+00 9.000E+00

ITER-FEAT (B2-Eirene simulation)



Z MIN = 0.00E+00
MAX = 0.00E+00

Y MIN = -4.59E+02
MAX = -1.97E+02

X MIN = 4.07E+02
MAX = 6.98E+02

R A P S

Population escape factor

FZ-Juelich



-1.000E+00 -7.143E-01 -4.286E-01 -1.429E-01 1.429E-01 4.286E-01 7.143E-01 1.000E+00

Population escape factors for ITER-FEAT conditions

As a proof for significance of radiative transport in ITER-FEAT conditions:

- plot of spatial variation of population escape factor
- white region: opt. thin equil. (usual assumption, no absorption)
- yellow region: opt. thick equil. (absorption = emission)
- black region: neg. values (net absorption locally)

Note: this first iteration gives an indication of possible absorption/emission behaviour in further B2-EIRENE iterations

Next steps

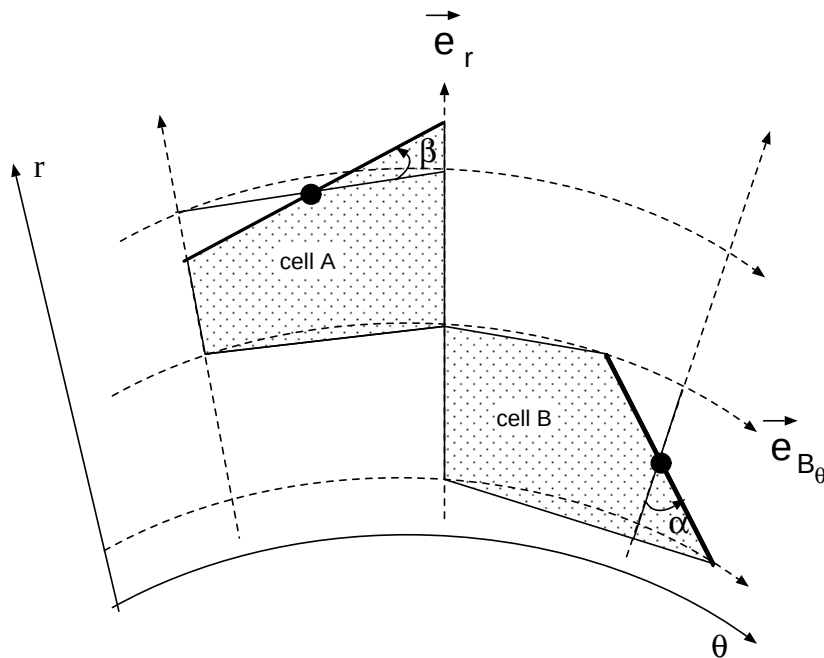
This needs to be assessed consistently with the plasma.

Various coupling methods between EIRENE and plasma codes are available.

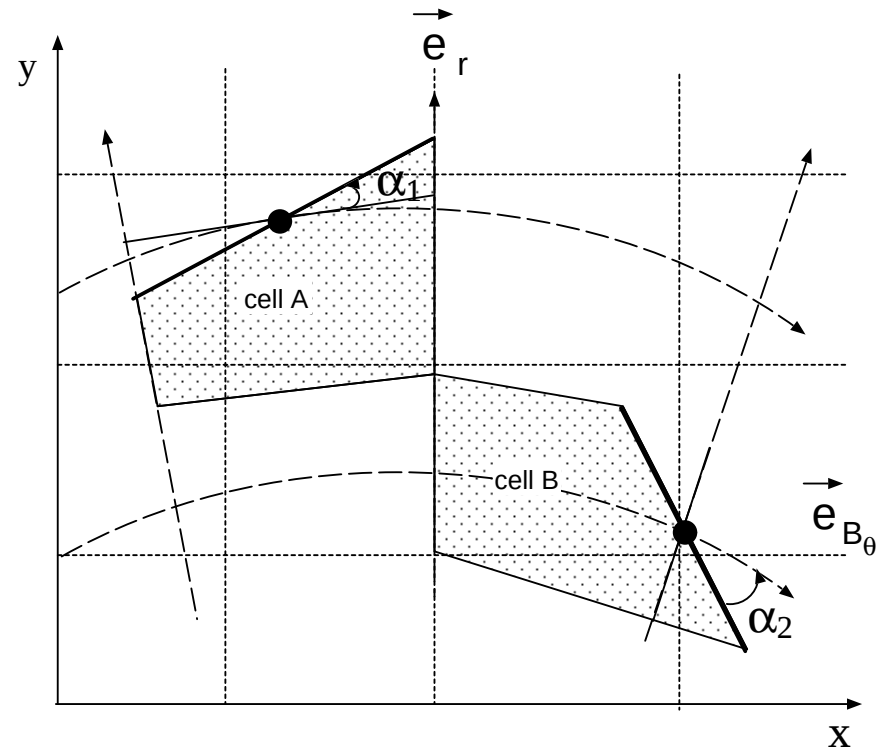
(See chapter 4 of EIRENE manual: EIRCOP)

Coupling plasma fluids to neutral kinetics, essentials:

Plasma fluids live in flux-surface co-ordinates



Neutral particles live in Cartesian co-ordinates



Coupling plasma fluids to neutral kinetics, essentials:

Plasma fluxes at boundaries have to match neutral recycling fluxes at the same boundaries **perfectly**.

(Note: Helium pumping determines the fate of fusion reactors, but is a 1% effect in recycling fluxes).

B2-EIRENE:

- B2 flow field given with respect to B-field: co-ordinate transformation o.k
- B2 fluxes given (and printed) at cell boundaries: flux conservation o.k
- all terms transferred to permit balance checks on both sides o.k.

B2.5-EIRENE

these 3 key feature are lost. Hence special attention is required.

Consistency of boundary conditions and flow field must be checked on a case to case basis. Report any mysteries, so that this can become more robust eventually.

B-field consistency check: trace ions also in EIRENE, with zero collisionality

• COUPLING EIRENE TO BRAAMS CODE: ASDEX UPGRADE SINGLE-NULL

SCALING FACTORS

FACT-X= 1.500E+02

FACT-Y= 1.500E+02

ORIGIN

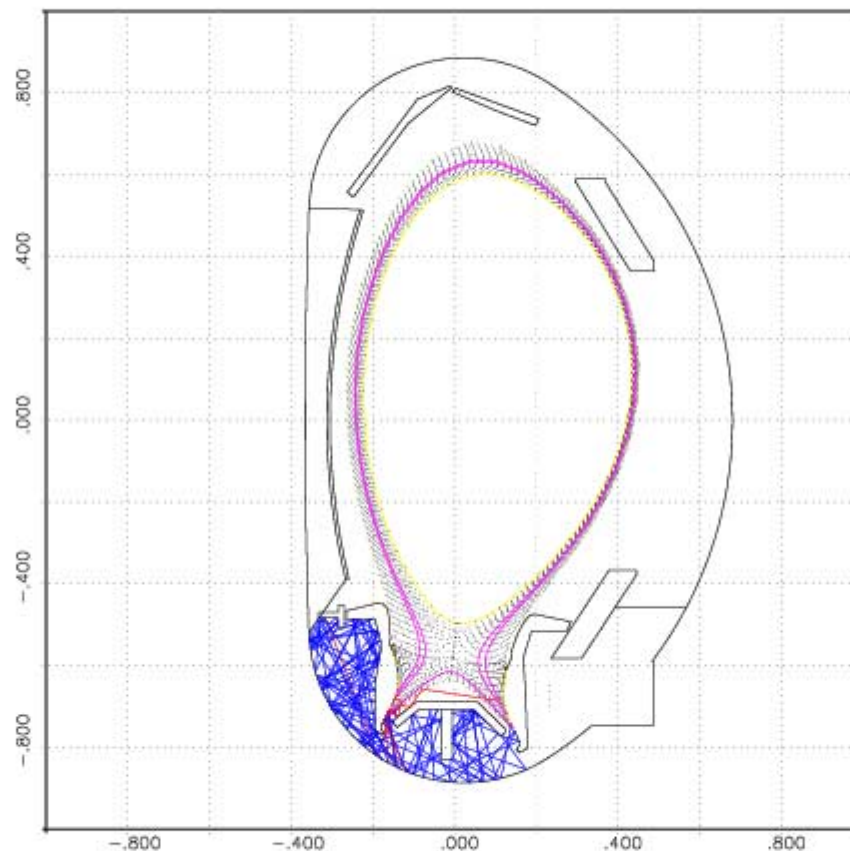
CH2X0= 1.500E+02

CH2Y0= 0.000E+00

PLOTTED AT

Z = 0.000E+00

- + LOCATE(1)
- ★ ELECTR. IMPACT(2)
- ELASTIC COLL.(3)
- × CHARGE EXCHANGE(4)
- FOKKER PLANCK(5)
- ▲ SURFACE(6)
- ◀ SPLITTING(7)
- ▼ RUSSIAN ROULETTE(8)
- PERIODICITY(9)
- RESTART:A, SPLT.(10)
- SAVE:COND. EXP.(11)
- RESTART:COND EXP(12)
- △ TIME LIMIT(13)
- ◇ GENERATION LIMIT(14)
- ★ FLUID LIMIT(15)
- ERROR DETECTED



H
H2(J=?)
H2+
H+
H

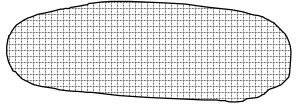
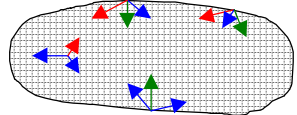
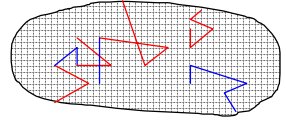
Next: the EIRENE manual

There are 14 distinct sections of an EIRENE input file

EIRENE

A general purpose linear transport
(Boltzmann equation) solver

Section of Eirene
Manual

DOMAIN	(discretised)	
SOURCE	of Objects	
OBJECTS	move on straight lines between Events	
EVENTS	Change direction and / or species at these events or their history ends at these events ("SINK for Objects")	

2,3

5,7

4,6

Output

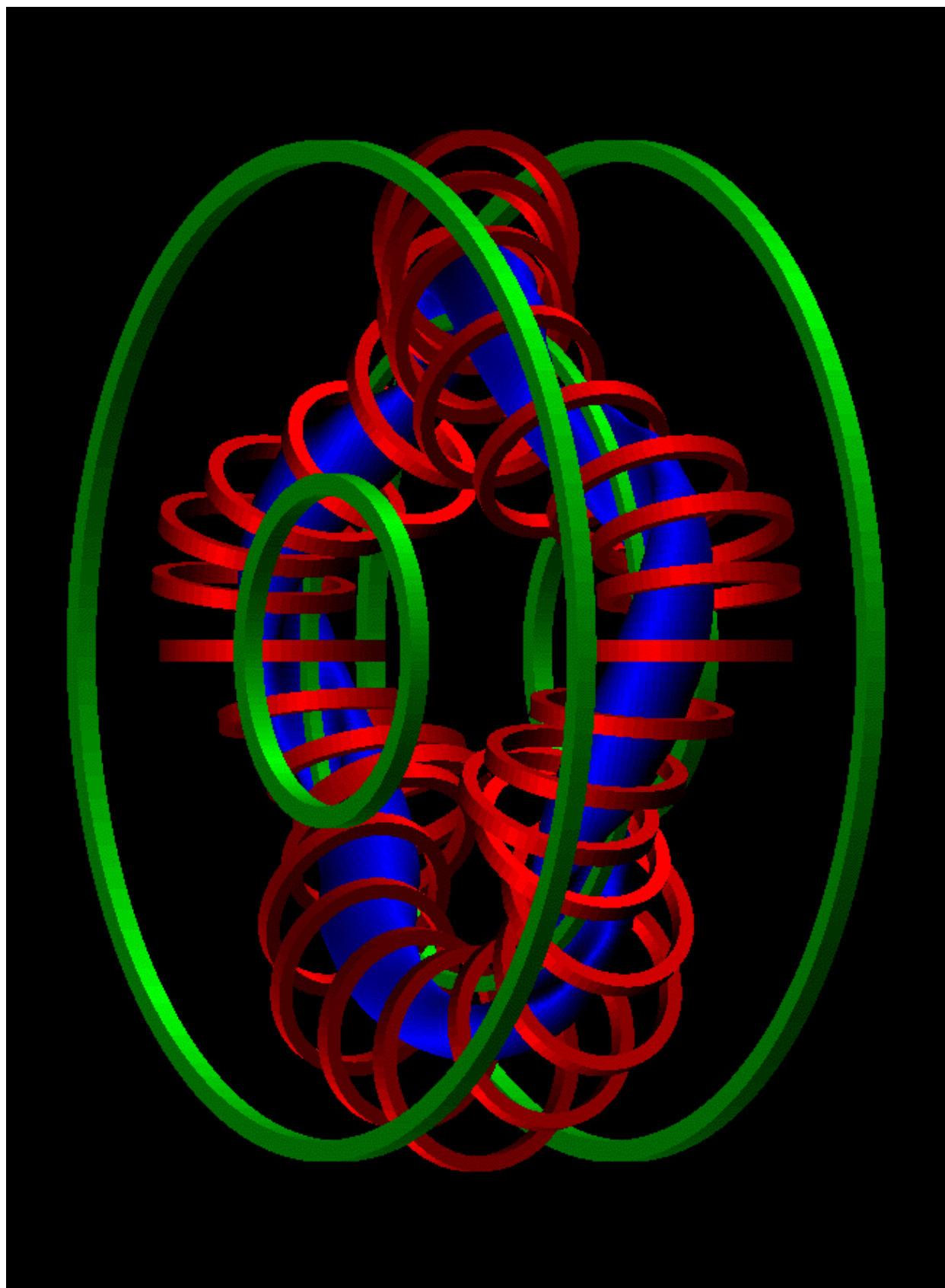
10,11,12

t-dependence

13

coupling

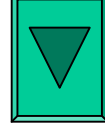
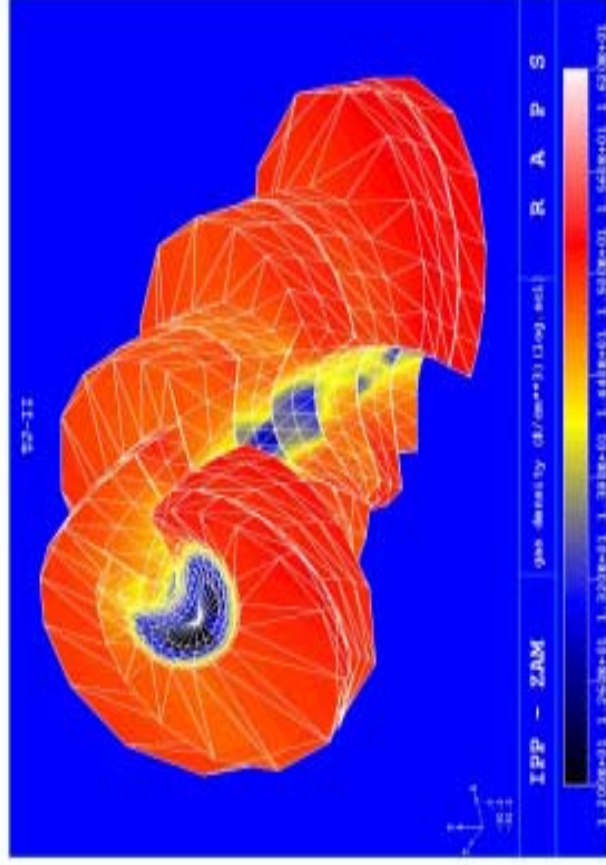
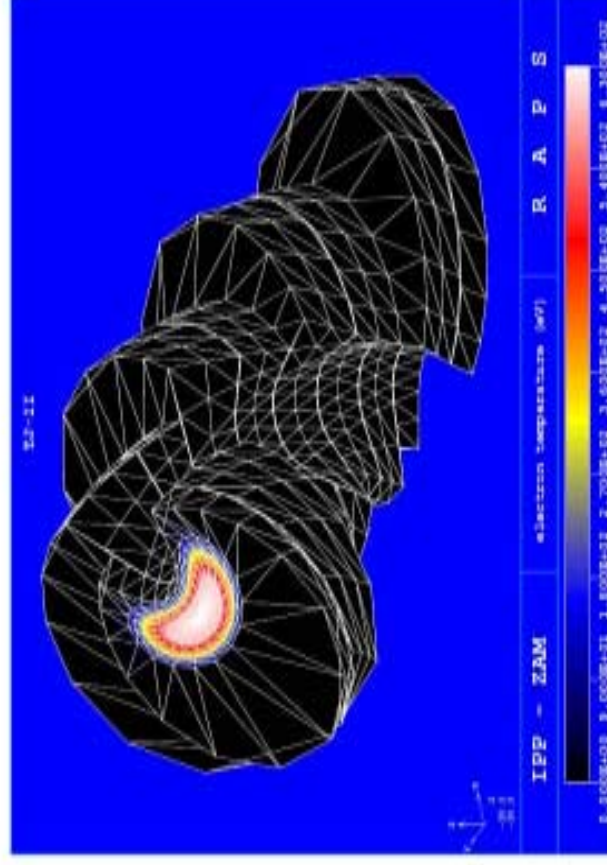
14



TJ-II: Flexible Heliac, Ciemat, Madrid



TJ-II: Flexible Helic, Ciemat, Madrid



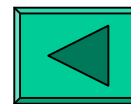
Transformation of phase-space

from velocity dependence (particle transport)

to

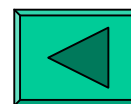
“one speed”, frequency dependence (photon
gas transport)

DONE



Lineshapes for emission and absorption due to

Natural broadening	done	
Stark broadening	done	very high electron dens.
Doppler broadening	done	Fusion plasmas
Pressure broadening	done	Lighting applications
Zeemann splitting	next	probably relevant



For simple test cases numerical solutions can be obtained